

IQOO HACKATHON 2026 · HYDERABAD CITY BATTLE

Chetaka

The on-device scam shield that follows
the call **into the payment app.**

TRACK · FINTECH & COMMERCE

FRAUD PREVENTION

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Digital-arrest scams turn one phone call into a UPI transfer within minutes.

2,41,537

Digital-arrest cases in India, 2022–2025

₹3,012 cr

Lost to those cases in the same period

46%

Of operations run from Cambodia, Myanmar and Laos

WHERE THE FRICTION IS

- 01

Protection looks at the wrong thing

Caller-ID apps judge who is calling. Scammers rotate clean numbers and pass every check.
- 02

Protection stops at the wrong moment

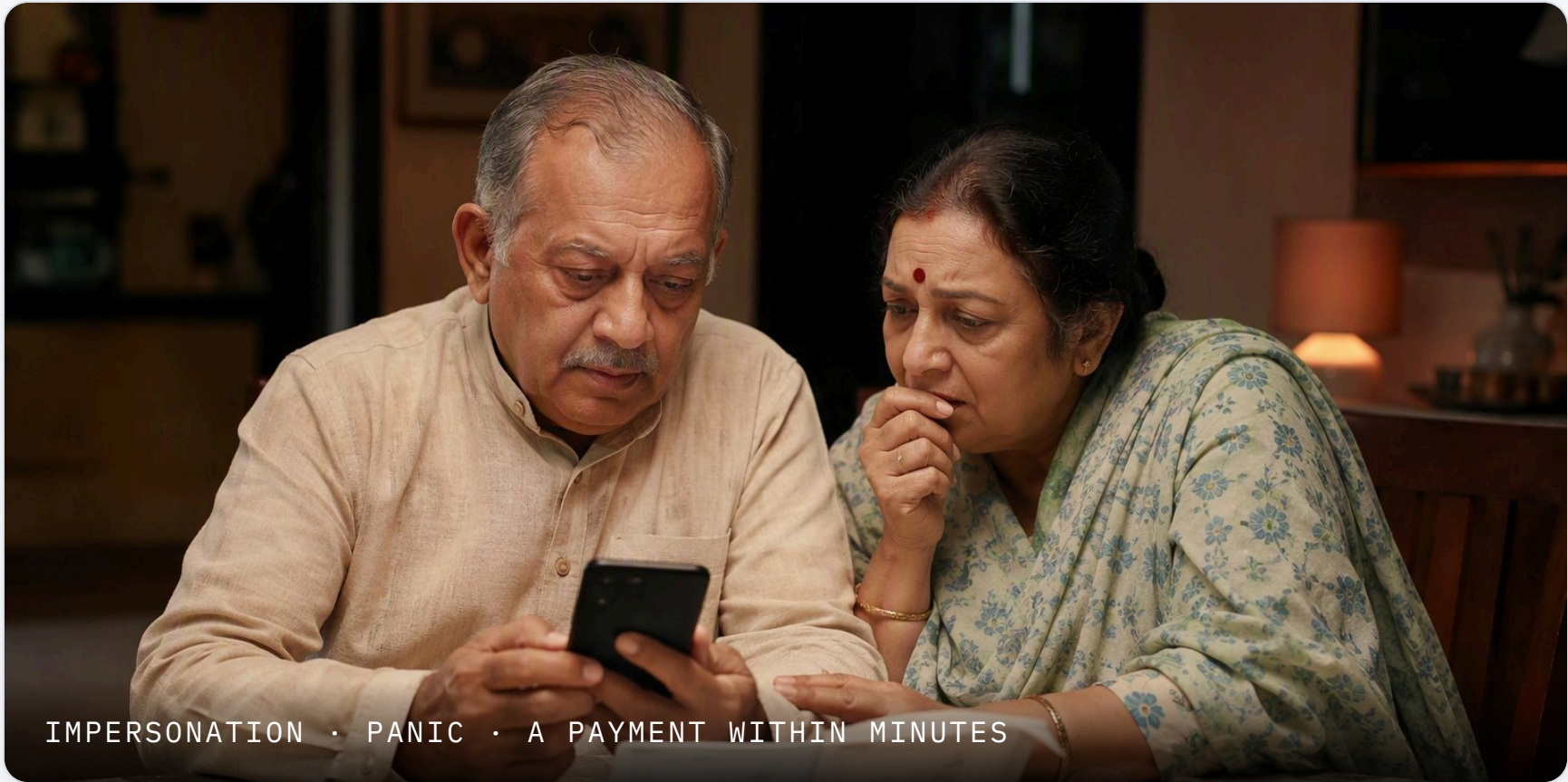
Call warnings end when the call ends. The money moves minutes later, inside the payment app.
- 03

Protection speaks the wrong language

Scam calls in Hyderabad mix Telugu, Hindi and English. Most tools understand English only.

WHO GETS HURT

Parents at home and first-time UPI users — people who trust authority, are isolated in the moment, and are pushed to act before they can check with anyone [4][6].



IMPERSONATION · PANIC · A PAYMENT WITHIN MINUTES

Source: MHA and I4C data as cited by Chetry (2026), Observer Research Foundation [7]

₹24 crore, 26 transfers, 73 days of fear.

BENGALURU · FEBRUARY–APRIL 2026

Lakshmi Ramamurthy, 74

Targeted after a property sale. Callers posing as CBI and Enforcement Directorate officers kept her under "digital arrest" from 10 February to 24 April 2026.

The call

Impersonated federal investigators; threatened arrest and asset seizure.

The pressure

Weeks of daily check-ins. She told nobody — isolation is the tactic.

The money

₹24 crore moved in 26 transactions to 23 mule accounts across 10 banks.

The end

Police intervened on 24 April, saving ₹3 crore. Six arrested; ₹5.5 crore recovered.

Source: The Federal, 25 May 2026 [9]

WHERE CHETAKA WOULD HAVE STEPPED IN

Transaction 1. An unknown number, a long call, then a banking app — the payment guard fires before the first rupee moves, and her son is alerted the same minute.

HOW MANY PEOPLE FACE THIS

2,41,537

Digital-arrest cases reported in India, 2022–2025

₹3,012 cr

Lost in those cases (Ministry of Home Affairs data)

₹1.25 lakh

Average loss per reported case (derived from the two MHA figures)

6,998

Indians rescued from scam compounds abroad, 2022–2025

WHO IS TARGETED

- Retirees holding one large, recently liquid sum.
- First-time digital payment users who trust authority.
- People alone at home, told to keep the call secret.
- **46%** of the operations run from Cambodia, Myanmar and Laos.

Figures as cited by Chetry (2026), Observer Research Foundation [7]

Warn during the call. Pause the payment after it.

One shield across the whole scam path — call, payment, family.

UNDERSTANDS

What the caller says, in Telugu, Hindi and English — on the phone.

INTERVENES

Before the money moves — the payment guard needs no call audio.

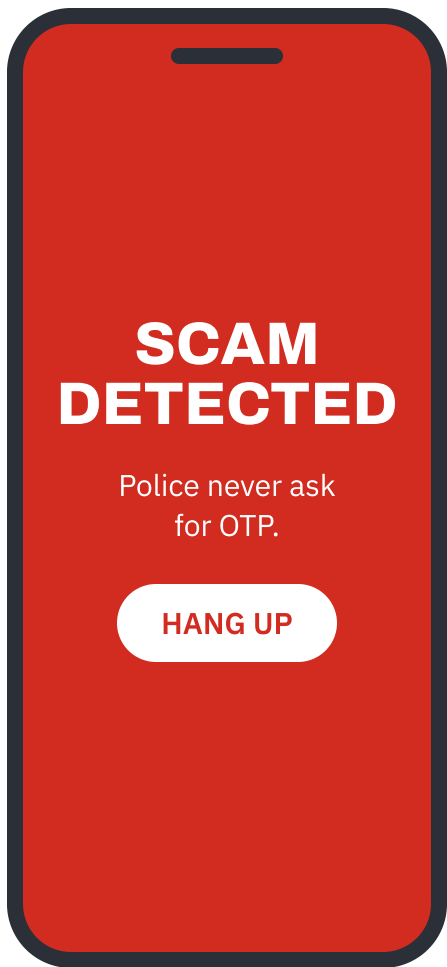
INVOLVES FAMILY

A trusted person is alerted at critical risk, in real time.

1 · DURING CALL

Warn mid-call

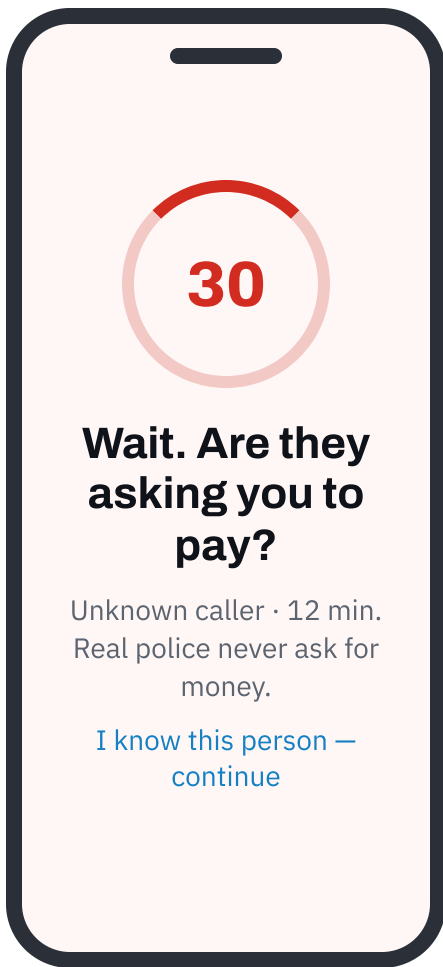
1. Unknown number calls.
2. Tap speaker — live listening needs speakerphone.
3. On-device transcript scored live.
4. High risk: full-screen alert + haptics.



2 · AFTER CALL

Pause the payment

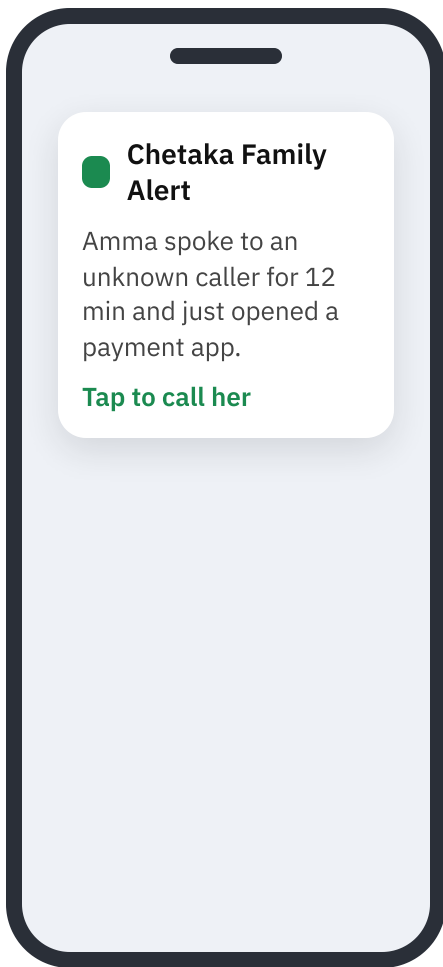
1. Call ends; length and caller kept.
2. UPI or bank app opens within 15 min.
3. 10-second pause screen.
4. One tap to skip if the payee is known.
5. Skip reason logged: known person, delivery, cab.



3 · FAMILY

Bring family in

1. Risk reaches critical.
2. Paired family phone notified.
3. One tap to call back.
4. Opt-in per contact; no audio shared.



Three AI tiers on the phone. Zero cloud AI calls.

PHONE SIGNALS

Voice · speaker only
MediaRecorder
VOICE_RECOGNITION

Telephony
Call state, caller, duration

App usage
UPI / bank app in foreground

TIER 0 · ALWAYS ON

CPU
Offline speech-to-text
Android SpeechRecognizer

CPU
Rules & phrases
Unknown caller, call length, multilingual scam phrases

TIER 1 · RISK ≥ 20

SNAPDRAGON NPU
Whisper (open-source)
Code-mixed Telugu · Hindi · English transcription [1]

SNAPDRAGON NPU
Scam classifier
Small transformer, LiteRT + Qualcomm NPU delegate

TIER 2 · RISK ≥ 60

Risk engine
Explainable 0–100 score with reasons

ON-DEVICE SLM STRETCH
Gemma 3 1B
Explains the scam tactic in the user’s language

PHONE ACTIONS

Alert + haptics
Overlay over the call

Payment pause
10-second guard, one-tap skip

Family alert
Paired phone notified

ON-DEVICE VS CLOUD

NEED	CHETAKA ON-DEVICE	CLOUD API
Private call audio	Never leaves phone	Uploaded
Mid-call latency	No network hop	Network-bound
Weak signal areas	Works offline	Fails
Running cost	Zero per call	Per request

CHETAKA

FALLBACK LADDER

- **NPU loads:** full tiers, latency shown on screen.
- **NPU fails:** models on GPU or CPU.
- **Models fail:** rules + phrases; all flows still work.

PHONE HARDWARE USED

Snapdragon NPU

Microphone

Haptics

Telephony

Overlays

Camera · QR check

HOW WE GET CALL AUDIO

No app gets the downlink stream. We record the room on speakerphone (MediaRecorder, VOICE_RECOGNITION).
AccessibilityService rejected — Play policy risk.
CallScreeningService is metadata only, so it feeds the guard, not the transcript.

FINTECH & COMMERCE · FRAUD PREVENTION

Native Android. Local AI runtime. No backend for AI.

MOBILE PLATFORM

Native Android

Direct access to telephony, overlays and the NPU.

Kotlin 2.0	Language
Jetpack Compose	UI and overlays
Coroutines + Flow	Real-time pipeline
Foreground Service	Listening during calls
minSdk 29 · targetSdk 35	Android 15 on the loaner
Android Studio + Gradle KTS	Build and profiling
JUnit + Compose UI test	Metrics harness

LOCAL AI RUNTIME

On-device models

Open-source models, quantised for the phone.

LiteRT + QNN delegate	Classifier on NPU
whisper.cpp · Whisper small	Speech-to-text
MediaPipe LLM Inference	Gemma 3 1B (stretch)
SpeechRecognizer	Offline fallback
int8 quantisation	Classifier ~25 MB · Whisper ~180 MB
Gemma 3 1B int4	~550 MB, loaded only at tier 2
All weights shipped in APK	No model download at runtime

DATA & SERVICES

Local first

Only the family alert touches the network. Office Kit mirrors the debug HUD — per-segment inference latency in ms and the live risk score — without interrupting the call.

Room + SQLCipher	Encrypted local store
TelephonyManager	Call state
UsageStatsManager	Payment app watch
Firebase Cloud Messaging	Family alert relay
iQOO Office Kit	Debug HUD mirrored to laptop

ANDROID PERMISSIONS — EACH TIED TO ONE FEATURE

READ_PHONE_STATE Call start and end	READ_CONTACTS Unknown-number check	RECORD_AUDIO Listening on speaker	PACKAGE_USAGE_STATS Payment app detection
SYSTEM_ALERT_WINDOW Alert and pause overlays	FOREGROUND_SERVICE_MICROPHONE Background listening	POST_NOTIFICATIONS Alerts on Android 13+	VIBRATE Haptic warning

Others protect the call. Chetaka protects the money.



Payment-app guard — our core innovation

Links a risky call to the payment app opened after it, and pauses before money moves. Content-independent, so it still works when the scam script fools a classifier [5]. Every skip is logged with a reason, so the guard learns.



Live vernacular detection

Telugu, Hindi and English code-mix, understood on-device mid-call.



Explainable risk score

Every alert names its reason: authority, OTP request, urgency, isolation.



Family Shield

Real-time alert to a trusted person. Opt-in per contact, no audio shared.



Private by design

No audio stored, no server AI, one-tap delete.

CAPABILITY	TRUECALLER	GOOGLE SCAM DETECT	BANK SMS ALERT	CHETAKA
Judges who is calling	Yes	Yes	No	Yes
Understands what is said	No	Varies	No	Yes
Telugu · Hindi · English mix	No	Limited	No	Yes
Guards payment app after call	No	No	No	Yes
Acts before money moves	Partly	Partly	After	Yes
Real-time family alert	No	No	No	Yes
Fully on-device AI	Varies	Varies	—	Yes

RESEARCH GAP

Voice-phishing classifiers judge the transcript alone, and LLM-written scam scripts cut their accuracy by more than 30% [5]. Chetaka pairs AI detection with a second check at the payment step.

A working prototype on the iQOO phone in 30 hours.

MINIMUM VIABLE PROTOTYPE

Must work in the demo

- Unknown-call detection and call log
- Payment-app guard with pause + skip log
- Live transcription + scam scoring
- In-call alert with haptics
- Family alert on a second phone

Stretch goals

- Scam classifier on the NPU, latency on screen
- Gemma 3 1B explanation in Telugu / Hindi
- Camera UPI-QR check

30-HOUR BUILD ORDER

0-5 h	Call monitor, contact check, local store
5-9 h	App watcher + payment pause — core innovation first
9-14 h	Listening service, speech-to-text, risk engine
14-17 h	In-call alert and haptics
17-20 h	Family alert and pairing
20-24 h	NPU classifier, Whisper, SLM — stretch
24-30 h	Test set, polish, demo rehearsal

LIVE DEMO · 3 PHONES

1. Phone B calls Phone A as "police".
2. Chetaka alerts mid-call on Phone A.
3. Repeat without speaker; Phone A opens a UPI app.
4. Pause screen holds the payment; skip shown.
5. Phone C gets the family alert; debug HUD mirrored to the companion screen via Office Kit.

HONEST EVALUATION

Measured live on scripted calls, both languages, reported as-is. Targets: **recall ≥ 0.80, false positives ≤ 10%, alert under 3 s**. A wrong block costs more than a missed one, so the false-positive target is the hard one.

20 scam calls	20 normal calls	3 metrics
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Recall · false-positive rate · median alert latency

Buildable today. Useful to millions of families.

01 Feasibility

- Every signal uses **public Android APIs**.
- All AI models are **open-source**.
- The core guard **needs no AI model**.
- **Fallback ladder** keeps flows working.

02 Benefits

- Stops payment **before the loss**.
- Payment guard works **with speakerphone off**.
- **Private**: nothing leaves the phone.
- **Zero cloud cost** at any scale.

04 Why Chetaka should be selected

- **Practical utility**: acts at the exact moment money is lost.
- **Novel**: the only approach guarding call-to-payment end to end.
- **Phone-first**: voice, telephony, haptics and NPU AI on-device.
- **Feasible**: model-free core first, AI tiers layered on top.

03 Target users



- **Parents at home** receiving impersonation calls.
- **First-time UPI users** in tier-2, tier-3 and rural India.
- **Adult children** protecting family remotely.
- **Partners**: phone makers, banks, telcos, cyber cells.

Research and data behind Chetaka.

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[5]

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[Open source](#)

Thank you

CHETAKA · THE SCAM SHIELD THAT FOLLOWS THE CALL